

Solar Superstorms: Planning for an Internet Apocalypse

The paper attempts to quantify the impact of black swan events, particularly solar superstorms, on global internet infrastructure and proposes general guidelines to brace for such events. A single day of global internet outage can result in devastating economical impact. Geomagnetically induced current resulting from the magnetized particles ejected by the Sun, through electromagnetic induction can wreak havoc on long distance cables, which are the backbone of the internet. Internet was developed during the low activity period of the sun, which is now ending. Hence, we have limited understanding of the effects of solar storms on the internet infrastructure. The paper discusses four major areas, namely, the magnitude of the threat, its impact on longhaul and submarine cables, quantifying its impact using real world datasets and the steps to tackle such an event.

Solar superstorms can destroy satellites, and although fiber optic cables may be immune to the storms, its repeaters are not. The paper discusses the impact of such events in four key areas. Internet infrastructure distribution is skewed towards higher latitudes, which are more susceptible to the impacts of solar superstorms. Submarine cables, owing to their length, are more prone to damage in such events than land cables. Intra-region connectivity would sustain but inter-region connectivity, specifically for regions separated by sea, would not. Likewise, Google's datacenters have better resilience than Facebook as they are not concentrated in one region. Lastly, DNS root servers are globally distributed and hence are less vulnerable.

The paper presents some preventive measures that can be taken to prepare for the next solar superstorm. To this end, the first proposed suggestion is to electrically isolate submarine cables connecting higher latitudes to lower ones to prevent cascading failures. The author also suggests the development of standardized tests for measuring end to end resilience. In case of such an event, we may get between 13 hours to 3 days before CME reaches the Earth. This time can be utilized to shut down electrical components to minimize damage. The author also recommends development of new seamless protocols that combine all available modes of communication.

The paper presents insights on a phenomenon that not only are we ill-prepared for, but also mostly unaware of, as the last such event happened over a century ago. Modern day internet applications often rely on the high speed internet available today. Even if such an event results only in greatly reduced internet speeds, instead of complete outage, it would still cause a lot of problems. This recent LinkedIn post demonstrates our reliance on high speed internet.

An area of improvement for the paper would be to include the distribution of Amazon and Microsoft's datacenters, in addition to Google and Facebook that the paper mentions. AWS has the highest cloud market share, followed by GCP and Azure, which means that most of the internet is hosted on these services. Lack of resilience for such events in the infrastructure of these companies would have impacts beyond these companies' own services. Lastly, some interesting areas to explore would be to study the role of services like Tor, or the effects of blocking of peer to peer traffic by various governments during such events.

Facebook outage: what went wrong and why did it take so long to fix after social platform went down?

This short article complements the paper summarized above. The cause of outage, as described in the article, was a configuration change to the backbone routers which connect company's datacenters. This configuration change had a cascading effect, which brought all of the company's services down for several hours. The article describes that Facebook deployed a series of updates, which essentially misconfigured BGP, resulting in the outage. The outage took several hours to fix as the Facebook employees were unable to access their own communications platform, hosted in the same place. Fixing the problem required manual reset to the servers. This event resulted in a 4.9% drop in Facebook's share price.

The article shows how a minor, relatively easily fixable human error caused a multitude of problems for Facebook. Not only did it cause several hours of outage, but also had a huge financial impact for the company. In the event of a global internet outage, which can be caused by Solar Superstorms, the financial impact would span across multiple businesses and could amount to billions of dollars per day. The services that exist solely due to internet may cease to exist and everything that internet has replaced in the modern world would suffer. We are accustomed to mobile banking, debts and loan records are digitized and many services do not exist outside the internet. A large scale internet outage would impact all areas of life and could cause stock markets to collapse.